

Fermi and Luttinger Arcs: Two Concepts, Realized on One Surface

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We present an analytically solvable model for correlated electrons, which is able to capture the major Fermi surface modifications occurring in both hole- and electron-doped cuprates as a function of doping. The proposed Hamiltonian qualitatively reproduces the results of numerically demanding many-body calculations, here obtained using the dynamical vertex approximation. Our analytical theory provides a transparent description of a precise mechanism, capable of driving the formation of disconnected segments along the Fermi surface (the highly debated “Fermi arcs”), as well as the opening of a pseudogap in hole and electron doping. This occurs through a specific mechanism: The electronic states on the Fermi arcs remain intact, while the Fermi surface part where the gap opens transforms into a Luttinger arc.

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Introduction—The emergence of isolated arcs on the Fermi surface (FS) of strongly correlated systems, such as those characterizing the famous pseudogap regime [1,2] of high-temperature superconducting cuprates [3], has been the object of an incessant and quite vivid theoretical debate in the last thirty-five years.

In the specific case of the cuprates, the presence of Fermi arcs is suggested not only by the evolution of the ARPES spectra as a function of temperature and doping [4–14], but also by the persistence of clear Fermi liquid features in the qualitative behavior of transport experiments performed in the whole normal phase of these compounds [15,16].

At the same time, the emergence of Fermi arcs remains quite puzzling from a theoretical point of view: On the one hand, they would be forbidden in conventional band theory [17,18] (where bands are continuous and Fermi surfaces closed), and on the other hand, the complexity associated with the explicit inclusion of electronic correlations hinders the identification of a transparent mechanism explaining their formation.

In the latter respect, the quest for explaining the enigmatic properties of the celebrated pseudogap phase in the cuprates has led many groups to perform a huge number of theoretical calculations, ranging from phenomenological modeling of the pseudogap self-energy [19–26] to advanced quantum many-body calculations (mostly of hole- or electron-doped Hubbard/Hubbard-like models) [27–34]. In several cases, the theoretical results have provided spectral functions and transport properties compatible with the experimental observations in the cuprates [35,36]. Further, the application of the fluctuations diagnostics approach [37–43] has highlighted the predominant role played by strong spin fluctuations [44–48] in driving the

outcomes of quantum many-body calculations in the interaction and doping regimes relevant to cuprates [49].

However, neither the phenomenological approaches nor the most advanced numerical studies have so far provided an intuitive and physically intelligible picture of the observed phenomena.

In this paper, we approach the problem from a different perspective: We introduce an *exactly solvable* many-electron model Hamiltonian on a two-dimensional (2D) lattice. This Hamiltonian is able to describe—depending on the parameter regimes considered—the formation of Fermi arcs in different sectors of the FS and to capture the qualitative differences in the spectral properties between electron- and hole-doped compounds, including the pseudogap properties. It is worth underlining already here that, while the conception of our Hamiltonian is—to a given extent—inspired by the Hatsugai-Kohmoto model (HKM) [50], which has recently attracted an increasing interest both for its physical content and for its formal properties [51–64], its key ingredient presents a fundamental difference: The electronic interaction of our model is *not* purely diagonal in momentum space, as it is in the HKM, but rather, it couples all momenta differing by the transfer momentum of $\mathbf{Q} = (\pi, \pi)$ associated with antiferromagnetic (AF) spin fluctuations. As we demonstrate in the following, this feature enables our model, differently from the HKM, to capture the fundamental momentum-dependent modifications of the FS occurring in 2D correlated electron systems as a function of hole/electron doping, in reasonable agreement with those we obtain by means of cutting-edge quantum many-body schemes, such as the dynamical vertex approximation (D Γ A) [26,65,66]. In this context, the invaluable insight gained by mastering an analytically solvable model provides a new route to clarify the hotly debated questions of the emergence of Fermi arcs, of the mechanisms controlling

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their evolution, and consequently, of the intrinsic nature of the pseudogap phase.

Model Hamiltonian—We introduce here our model Hamiltonian for a square lattice, which (i) allows for a general exact analytical solution and (ii) displays Fermi arcs, defined as disconnected parts of the FS. It reads

$$H = \sum_{\mathbf{k}\sigma} \left[(\epsilon_{\mathbf{k}} - \mu) \hat{n}_{\mathbf{k}\sigma} + \frac{\mathcal{V}}{2} \hat{n}_{\mathbf{k}\sigma} \hat{n}_{\mathbf{k}+\mathbf{Q}-\sigma} \right], \quad (1)$$

where μ is the chemical potential, $\hat{n}_{\mathbf{k}\sigma} = \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma}$ is the occupation operator, and $\mathcal{V}$ is the interaction strength that couples the momenta $\mathbf{k}$ and $\mathbf{k} + \mathbf{Q}$, with $\mathbf{Q} = (\pi, \pi)$ describing the characteristic momentum of the coupling. For the dispersion relation $\epsilon_{\mathbf{k}}$, we consider

$$\epsilon_{\mathbf{k}} = -2t(\cos(k_x) + \cos(k_y)) - 4t' \cos(k_x) \cos(k_y) - 2t''(\cos(2k_x) + \cos(2k_y)), \quad (2)$$

where we use $t' = -0.2$ and $t'' = 0.1$ (while $t = 1$ sets the unit of energy) consistent with typical values used in low-energy tight-binding models for cuprates [26,67]. See the Supplemental Material [68] for a brief discussion regarding the role of the sign of the hopping parameters.

Relation to the Hatsugai-Kohmoto model—As mentioned in the Introduction, our Hamiltonian can be linked to the Hatsugai-Kohmoto model (HKM) [50], which can be retrieved by setting $\mathbf{Q} = 0$ in Eq. (1) and also allows for an exact solution. We recall here that the HKM can be mapped to a set of decoupled atomic limit models [$H_{AL} = U \hat{n}_\uparrow \hat{n}_\downarrow - \mu_{\text{eff}}(\hat{n}_\uparrow + \hat{n}_\downarrow)$] in the absence of a magnetic field, one for each $\mathbf{k}$ point, whose difference is only in the value of their effective chemical potentials $\mu_{\text{eff}} = \mu - \epsilon_{\mathbf{k}}$. This property, which renders the model solvable, also allows for a description of the appearance of **Luttinger zeros at the Fermi energy and of Mott-like insulating phases** [50,51,76,77]. At the same time, we note that the complete decoupling of the HKM into decoupled H_{AL} models severely restricts the possibility to describe second-order phase transitions. Further, recent work [78] has pointed out possible intrinsic inconsistencies in the linear response expressions of the HKM.

The crucial difference of the HKM with respect to our model arises from the nonzero value of $\mathbf{Q} = (\pi, \pi)$ in Eq. (1). This captures the coupling of electrons at different momenta (absent in the momentum-diagonal HKM), mediated by AF fluctuations, eventually allowing the emergence of momentum-selective behaviors. To a given extent, Eq. (1) can be viewed as a set of atomic limit models in momentum space in the presence of a *magnetic field* $h_{\mathbf{k}} = (\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}+\mathbf{Q}})/2$ and chemical potential $\tilde{\mu}_{\mathbf{k}} = \mu - (\epsilon_{\mathbf{k}} + \epsilon_{\mathbf{k}+\mathbf{Q}})/2$ acting on two states of opposite spin at different momenta (i.e., $\mathbf{k}\uparrow, \mathbf{k} + \mathbf{Q}\downarrow$) (see also Fig. S3 in the Supplemental Material [68]).

Analytic solution—To analyze the evolution of the FS, and in particular the possible emergence of disconnected sections, we start by computing the single-particle Green's function as a function of the Matsubara frequency ν_n : $G_{\mathbf{k}\sigma}(i\nu_n) \equiv -\int_0^\beta d\tau \langle \hat{c}_{\mathbf{k}\sigma}(\tau) \hat{c}_{\mathbf{k}\sigma}^\dagger(0) \rangle e^{i\nu_n \tau}$.

For the model proposed in Eq. (1), $G_{\mathbf{k}\sigma}$ can be analytically computed directly on the real-frequency axis

$$G_{\mathbf{k}}(\omega) = \frac{1 - n_{\mathbf{k}+\mathbf{Q}}}{\omega + \mu - \epsilon_{\mathbf{k}} + i0^+} + \frac{n_{\mathbf{k}+\mathbf{Q}}}{\omega + \mu - \epsilon_{\mathbf{k}} - \mathcal{V} + i0^+}, \quad (3)$$

where $n_{\mathbf{k}+\mathbf{Q}} = \langle \hat{n}_{\mathbf{k}+\mathbf{Q}\sigma} \rangle$ (assuming no magnetic order) can be calculated according to Eq. (4) below. We note that the presence of $n_{\mathbf{k}+\mathbf{Q}}$ (instead of $n_{\mathbf{k}}$ as in the HKM) allows here for the coupling of different momenta, and hence for the possible emergence of momentum-selective features like Fermi arcs. Specifically, there are three different (momentum-dependent) possibilities for the Green's function: (i) For all momenta $\mathbf{k}$ at which $n_{\mathbf{k}+\mathbf{Q}} = 0$, we have the noninteracting Green's function. (ii) For all $\mathbf{k}$ at which $n_{\mathbf{k}+\mathbf{Q}} = 1$, only the second term in the rhs of Eq. (3) is nonzero, yielding the noninteracting solution but with a chemical potential shifted by $\mathcal{V}$. (iii) For all $\mathbf{k}$ at which $0 < n_{\mathbf{k}+\mathbf{Q}} < 1$, both branches of G in Eq. (3) are active, and the spectrum splits into two peaks with a gap in between.

For the four many-body states of each pair $\mathbf{k}\uparrow, \mathbf{k} + \mathbf{Q}\downarrow$ (see the Supplemental Material [68]), we can calculate

$$n_{\mathbf{k}+\mathbf{Q}} = \frac{e^{-\beta \xi_{\mathbf{k}+\mathbf{Q}}} + e^{-\beta(\xi_{\mathbf{k}} + \xi_{\mathbf{k}+\mathbf{Q}} + \mathcal{V})}}{1 + e^{-\beta \xi_{\mathbf{k}}} + e^{-\beta \xi_{\mathbf{k}+\mathbf{Q}}} + e^{-\beta(\xi_{\mathbf{k}} + \xi_{\mathbf{k}+\mathbf{Q}} + \mathcal{V})}}, \quad (4)$$

where $\beta = 1/T$ is the inverse temperature, and $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu$. This yields the evolution of the FS as a function of the chemical potential μ and the interaction strength $\mathcal{V}$ in the low- T limit ($\beta \rightarrow \infty$) shown in Fig. 1. Here, the color plot displays the corresponding filling, and the lines separate regions of qualitatively different Fermi surfaces. A representative quadrant of the corresponding FS is shown as an inset in each region.

This phase diagram can be understood further. Let us start with the arguably most interesting hole-doped case (V). For the part of the FS that lies inside the AF zone boundary (AFZB)—i.e., inside the AFZB lines $\pm k_x \pm k_y = \pi$ —the momentum $\mathbf{k}$ couples via $\mathbf{Q}$ to empty states with $n_{\mathbf{k}+\mathbf{Q}} = 0$. Thus, we obtain the noninteracting spectrum $A(\omega, \mathbf{k}) = \delta(\omega - \xi_{\mathbf{k}})$; the FS remains intact. Contrary to this, the part around the antinode [around $\mathbf{k} = (\pi, 0)$], which is outside of the AFZB, is coupled to filled states with $n_{\mathbf{k}+\mathbf{Q}} = 1$. Consequently, the spectrum $A(\omega, \mathbf{k}) = \delta(\omega - \xi_{\mathbf{k}} - \mathcal{V})$ is shifted to energies above the Fermi level. Hence, there is no FS for such momenta. **Instead, a Luttinger surface (LS) appears [79], defined by $G_{\mathbf{k}}(\omega = 0) = 0$ (i.e., with both the real and imaginary parts vanishing).** Resolving Eq. (3) for this condition yields

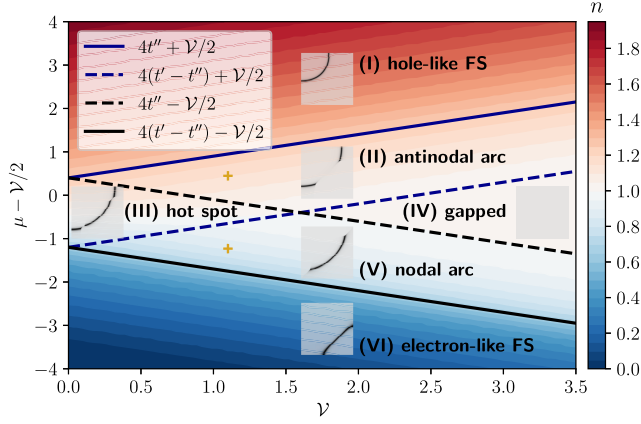


FIG. 1. Different phases of disconnected Fermi surfaces realized in the model Hamiltonian for $T \rightarrow 0$ ($\beta = 1/T = 10^6/t$). The color scale in the background displays the electron density n as a function of the chemical potential μ and interaction strength V . The black and blue solid and dashed lines, which depend on the interaction and the dispersion ϵ_k ($t = 1$, $t' = -0.2$, $t'' = 0.1$) divide the parameter space into six regions of qualitatively different spectral functions: (I) hole-like Fermi surface, (II) antinodal Fermi arc, (III) pseudogap on the hot spots, (IV) completely gapped, (V) nodal Fermi arc, and (VI) electron-like Fermi surface. An exemplary momentum-resolved spectral function of each region is displayed in the small insets. The gold “+” symbols indicate the values of V and μ used in Figs. 2 and 4.

$\xi_k = \epsilon_k - \mu = (n_{k+Q} - 1)V$, which for $\beta \rightarrow \infty$ becomes fulfilled close to the AFZB.

The result is a disconnected FS—or more precisely, a transition of a FS into a LS at the AFZB. This insight of an exactly solvable model is crucial to understanding Fermi arcs: It shows that Fermi surfaces do not need to be connected, but can display a direct transition into Luttinger arcs, which are invisible to ARPES experiments. The underlying combined surface (LS and FS) defined by $\Re G_k(\omega = 0) = 0$ remains continuous and connected. However, it splits into Fermi arcs [poles of $\Im G_k(\omega = 0)$] and Luttinger arcs [zeros of $\Im G_k(\omega = 0)$]; see also the Supplemental Material [68]. The location of the Fermi arcs in our model is qualitatively consistent with ARPES experiments in cuprates [4–14]. Further, the evolution of the FS/LS as a function of doping is directly reflected in the corresponding low- T behavior of thermodynamic properties, such as the electronic specific heat for $T \rightarrow 0$ (see the Supplemental Material [68]).

For the electron-doped case (II), the scenario gets reversed with respect to the AFZB; see panels (a) and (c) of Fig. 2. This again agrees with the experimental observation in cuprates [80–82].

It is also worth stressing here that the close proximity of Luttinger and Fermi arcs, which characterizes the pseudogap regimes (II) and (V), may lead one to suspect a mismatch in the Luttinger count [83] $n = \sum_{k\sigma} \Theta(\Re G_k(\omega = 0))$ similar to the one reported

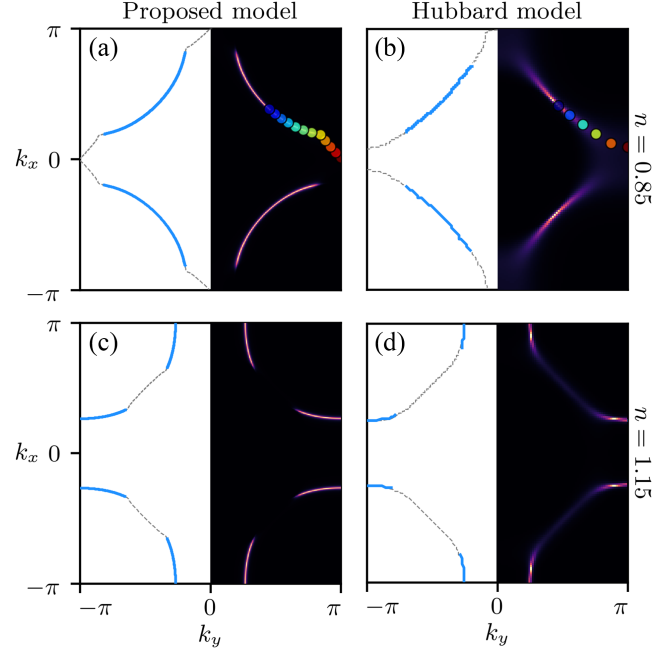


FIG. 2. Fermi surface (blue line) and Luttinger surfaces (gray line) computed for the proposed model at $V = 1.1t$ (left column) and, by means of DGA, for the Hubbard model at $U = 8t$ (right column) at hole (top row) and electron doping (bottom row) at $\beta = 12.5/t$. The left half of the Brillouin zone displays the Fermi surface (blue solid line) and Luttinger surface (gray dashed line), while the right half shows the spectral function at zero frequency [$A(\omega = 0, k)$]. Colored dots in the top row mark locations for the cuts in Fig. 4.

in Mott insulators [84,85]. In fact, for momenta close to the Luttinger surface, one does find that $(1/2\pi) \int_{-\infty}^{\infty} d\nu e^{i\nu 0^+} G_k(i\nu) [\partial \Sigma_k(i\nu) / \partial i\nu] \neq 0$ [86,87], with $\Sigma_k(i\nu)$ being the corresponding self-energy. However, for $T \rightarrow 0$, where Luttinger’s theorem applies [83,88], such violations remain confined on one-dimensional paths in the Brillouin zone (the AFZB) and hence vanish in the corresponding momentum integration.

Let us also briefly discuss the other physical regimes shown in Fig. 1. In particular, since the Fermi surface becomes disconnected only at the AF zone boundary, all values of μ , V for which $\xi_k = 0$ and $\xi_k + V = 0$ do not cross the AF zone boundary will show only a regular continuous hole- (I) or electron-like (VI) Fermi surface or a completely gapped half-filled system (IV). From this restriction, we can deduce the two regions (II), $4(t' - t'') + V < \mu < 4t'' + V$, and (V), $4(t' - t'') < \mu < 4t''$, where the (anti)nodal arcs are present. When the two regions intersect, both arcs are visible (III), similar to the so-called “hotspots” observed in experiment [81,89].

Connection to the Hubbard model—After illustrating the mechanisms through which Eq. (3) displays Fermi arcs, an important question to be addressed is how our findings connect to the physics described by the microscopical

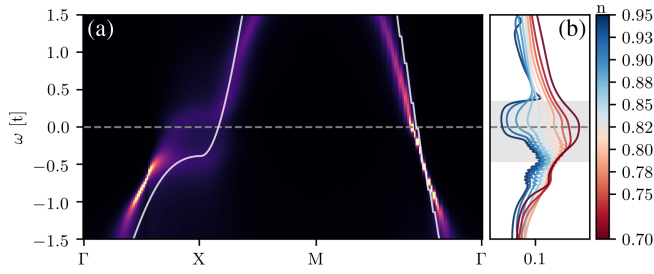


FIG. 3. (a) Spectral function $A(\omega, k)$ along a high-symmetry path $\Gamma = (0, 0)$, $X = (\pi, 0)$, and $M = (\pi, \pi)$, for the analytically continued DGA solution of the Hubbard model at $n = 0.85$ and $U = 8t$. The white line displays the noninteracting band dispersion for the same filling. Visible is the dichotomy of the spectrum between the antinode, which is gapped, and the node, which is not. This is directly reflected in the dip at the Fermi energy of the momentum-integrated spectral function $A(\omega)$, displayed in panel (b) for a range of dopings. As expected and observed in cuprates [7], the pseudogap is stronger at lower doping, and we observe no suppression for $n \leq 0.7$.

Hamiltonian typically adopted for describing the low-energy properties of cuprates—namely, the two-dimensional Hubbard model (HM):

$$H = \sum_{k\sigma} (\epsilon_k - \mu) \hat{n}_{k\sigma} + \frac{U}{2} \sum_{i\sigma} \hat{n}_{i\sigma} \hat{n}_{i-\sigma}, \quad (5)$$

where ϵ_k is the same dispersion as in Eq. (1) and $U = 8t$ is the strength of the on-site electrostatic repulsion. It is well known that any reliable calculation of the spectral properties of the HM requires advanced numerical many-body approaches [28]. Here, we resort to the dynamical vertex approximation (DGA) [26,65,66], which is a diagrammatic extension [90,91] of the dynamical mean-field theory [92–94], capable of treating nonlocal spatial correlations [95] even in the intermediate- to strong-coupling regimes of cuprate physics [96]. In fact, while not exact, DGA compares well to numerically exact quantum Monte Carlo, where such solutions are available [29], and it was recently also successfully applied to the study of nickelate superconductors [97].

We summarize our DGA results for the FS of the HM, calculated at $\beta = 12.5/t$ (for lower temperatures and computational details, see the Supplemental Material [68]), at both hole and electron doping in the right panels of Fig. 2. Here, clear pseudogap features are visible in terms of a significant suppression of the Fermi-level spectral intensity at the antinode (node) for the hole-(electron)-doped case, while the FS around the node (antinode) remains quite sharp and well defined. The overall behavior can be more precisely understood by looking at the full frequency dependence of the spectral function $A(\omega, k)$ along a high-symmetry path across the BZ, for a specific case [$n = 0.85$, see Fig. 3(a)]. Evidently, this interaction-driven momentum-selective behavior at

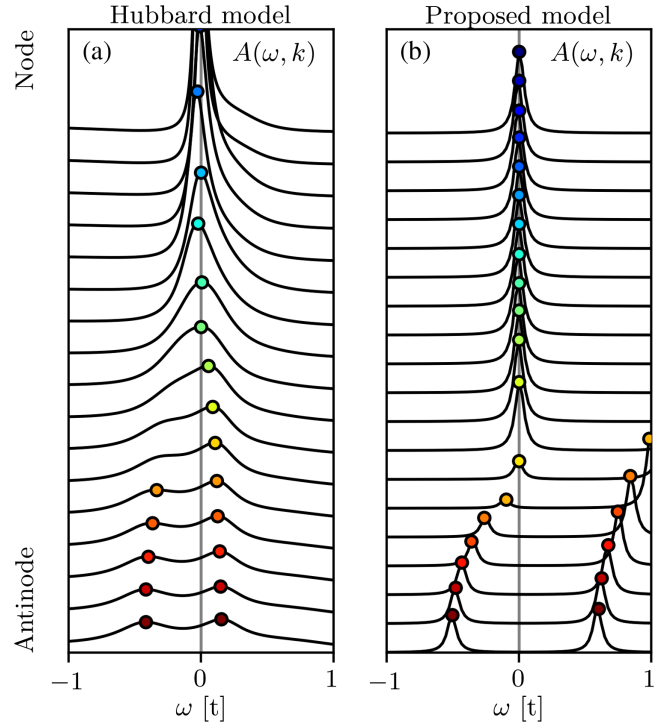


FIG. 4. Structure of the spectral function along the Fermi-Luttinger surface, for the momenta indicated by the same color in Fig. 2. Both DGA and the model are shown for the same filling $n = 0.85$ and inverse temperature $\beta = 12.5/t$. (a) Hubbard model using DGA at $U = 8t$. (b) Exact solution of the model in Eq. (1) at $V = 1.1t$.

the Fermi energy well resembles that observed in our proposed model in Eq. (1). In particular, both our model and the DGA solution of the HM yield the same evolution of Fermi (blue lines) in Luttinger arcs (gray dashed), as displayed in Fig. 2. At the same time, we note that the Fermi arcs computed in DGA do not end at the AFZB, but are more gradually smeared out, directly reflecting the momentum evolution of the (imaginary part of) DGA self-energy, which is smallest at the node (antinode) and increases along the arc, thus resulting in a broader spectrum. This feature is directly reflected in the doping evolution of the k -integrated spectral function $A(\omega)$, shown in Fig. 3(b) for the hole-doped case.

In order to visualize the full frequency dependence of the spectrum along the arc, we plot $A(\omega, k)$ in Fig. 4 at points located on the arc [marked by colored dots in Figs. 2(a) and 2(b)] for the Hubbard model and Eq. (3), respectively. The momentum cuts start at the node (top), where a single peak, reminiscent of a correlated metal, is visible. This peak becomes broader toward the AFZB, where it starts to split into two, which also marks the onset of the LS. The differences between the DGA and our analytical spectra can be easily rationalized in terms of the simplified interaction in Eq. (1), which preserves $\hat{n}_k$ as a good quantum number (and hence, the sharpness of the

spectra along the Fermi arcs) and is momentum independent (which is reflected in the fixed size of the interaction-driven gap). The overall evolution of the spectral functions, though, as well as that of the corresponding Fermi/Luttinger arcs, display an appreciable similarity. Here, one should note that, in spite of the good agreement of the spectral properties in the temperature range considered, capturing the overall T dependence of the AF fluctuations in D Γ A would require a T -dependent $\mathcal{V}(T)$ in Eq. (1). A fixed, T -independent $\mathcal{V}$ would model AF fluctuations essentially unaffected by temperature.

Conclusions—We have introduced a model Hamiltonian that, despite being *analytically solvable*, is able to capture, in the temperature range considered, the doping evolution of the Fermi surface as it is experimentally observed in hole/electron-doped cuprates *and* realized in cutting-edge numerical calculations for the doped Hubbard model, like the D Γ A employed here.

The advantage of having a full analytical description at our disposal has allowed us to rigorously investigate fundamental questions about the spectral properties of the normal phase of unconventional superconductors. In particular, our results (i) demonstrate that the emergence of disconnected parts of the FS (“Fermi arcs”) is possible, (ii) unveil the specific way in which this can be realized, and (iii) capture the differences between electron and hole doping. In particular, we have shown how Fermi arcs and Luttinger arcs, the latter ones invisible to ARPES, can reside on the same surface and *continually evolve* into one another, lifting the apparent necessity for closed Fermi surfaces and/or Fermi pockets with strongly varying spectral intensity. More precisely, in the presented model, the connection between FS and LS occurs at the k points where both cross the AFZB. The Fermi or Luttinger nature of the resulting segments (internal or external with respect to the AFZB) depends on the hole vs electron kind of doping, consistent with the differences of the spectra at the Fermi level characterizing electron- and hole-doped cuprates.

In summary, we have transparently interpreted the analytically calculated transformation of a Fermi surface into Fermi and Luttinger arcs in terms of the physics of an interaction-driven momentum-selective insulator, arising from the coupling of electrons whose momenta are linked by AF fluctuations.

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